Coding Workflow for RoO Restricted-Input Pairs

General coding rules for every preferential trade agreement (PTA). The definition of an input–output link is expansive: every pair **starts at DV = 1**, and each step turns to 0 only the cases it clearly excludes. Pairs are written **input** → **output** (A → B). The rules include the scope decisions of 19 October 2026 and the conventions of 28–30 October 2026 (Professor Zhang's questionnaire answers), with the points he settled on 1 November 2026; all are summarised in the last section. Results for individual PTAs are reported separately.

Decision rule

DV is the main variable (Professor Zhang, 1 November 2026). A pair is turned to 0 only on clear evidence that the input cannot be used to make the output: a step or convention below says 0, the direction is reversed, or the claimed route is physically or chemically impossible or too generic to name. A pair that cannot be settled stays 1 and is marked uncertain; it is never turned to 0 for being uncertain. Every pair that stays 1 falls into one of two levels:

Level	Meaning	Code
Known	A known production route of the output (or of the same kind of product) uses the input, or standard industrial practice does	DV = 1, uncertain = 0
Possible	Technically possible, but no known route confirms it	DV = 1, uncertain = 1
Invalid	Physically or chemically impossible, the reverse direction, or no specific route can be named	DV = 0

The three levels are Professor Zhang's overall principle of 30 October. The uncertain flag records which 1s are only possible; no variant of DV that counts only certain links is computed (1 November). Accuracy is reported for DV.

Scope. The conventions of 28–30 October apply to the pairs newly coded in PTA 6–10 and to all later agreements. Codes delivered for PTA 1–5 are not re-judged (Professor Zhang, 30 October). Pairs copied from them keep their codes (step 7a), including errors an audit finds in them, so a pair has the same code in every agreement. Two updates are pending for the next reissue of the PTA 1–5 files (Professor Zhang, 1 November): DV_strict is dropped from them, and under the blended-fabric convention the copied fabric pairs that share a minor fibre get uncertain = 1. In PTA 6–10 the conventions were applied by rule to whole classes; the answers that would raise single pairs from 0 to 1 and the distinctions between a certain 1 and a 1 with uncertainty were not re-judged pair by pair (see the results report). They apply in full to pairs not yet in the master list. Pairs already in it are copied as they are (step 7a), including PTA 6–10 pairs to which these answers were not applied (for example 842420 spray guns → 842410 fire extinguishers = 0). They are not re-reviewed; in particular, the convention on complete machines is not applied to pairs both coders coded 0 (Professor Zhang, 1 November).

Variables

Variable	Contents
pair_id, pta, hs_vintage	Identifiers
input_hs6, output_hs6, descriptions	The pair, in the agreement's own HS vintage; input_hs2002 and output_hs2002 give the common vintage (step 1)

DV	Final code; initialised to 1 for every pair
DV_excl_remelt	Robustness version of DV with direct remelting (route = recycle_direct) set to 0, per HS2002 pair
locked	1 if step 2 protects the pair at DV = 1
step	Step that turned the pair to 0 (3–6), or 2 for a certain same-good pair; blank otherwise
route	Reason a pair stays 1: same_good, forward, ingredient, component, reconstitute, mixture, recycle_waste, recycle_direct, feed, sibling_intermediate; unresolved (an open case copied from the ASEAN–Japan labels); ml_* (copied textile pairs kept at 1 by the model); none if turned to 0
uncertain, uncertain_reason	1 if the pair is kept at 1 at the "possible" level: the two AI coders disagree, a coder judged the link possible, a convention codes the case 1 with uncertainty, an open case is copied from the ASEAN–Japan labels, or the HS2002 pairs of a record differ (step 1); the reason is recorded
source	How the code was reached: exact_copy, ai_units, ai_sample, ai_concord, rule, rule_textile (ml only where the bands of step 7c are used)
policy	Decision or convention that changed the pair, if any
dv_ai1, dv_ai2, why_ai1, why_ai2	Each AI coder's code and one-line reason
ml_prob	Model probability (step 7 only)
concord_flag	1 if a code is not one-to-one with the common HS vintage, or not a valid code (step 1)

Step 1. Take in the RoO pairs and set DV = 1

- Read each agreement's product-specific rules and list every (restricted input, output) pair at HS6, in the agreement's own HS vintage.
- **Only HS6 inputs are coded.** Rows whose input is not an HS code — "WO" (wholly obtained) or "any" — are deleted.
- Map every pair to a common HS vintage (HS2002) with the WCO/UN correlation tables. A code that splits produces one pair per new code; flag non-one-to-one mappings as *concord_flag* = 1. Where two codes of the agreement fall into the same HS2002 code, the pair is coded on the agreement's own codes and descriptions. Any other record that maps to several HS2002 pairs takes the highest DV among them, with uncertain = 1 if they differ (unless input and output are the same code, which step 2 locks).
- A code that is not a valid subheading is mapped to every HS2002 subheading of its group and flagged.
- Remove duplicate pairs within each agreement and keep a pair-level master list across agreements (used in step 7).
- **Set DV = 1 for every pair.** Steps 3–6 can only turn a 1 into a 0; only a convention for a whole class of pairs (see "Decided by rule" under the two AI coders) can turn a 0 back into a 1.

The HS concordance will follow the project's HS conversion package once it is adopted; the rules above apply until then.

Step 2. Lock identical or same-good pairs at 1

Rule: set *locked* = 1 (*route* = same_good) when input and output are the same good or very similar: the same HS6 code, or the same species, material or variety differing only in processing stage, preservation state, packing or size of presentation, or a treatment the producer can add (roasting, decaffeinating, dyeing, printing, plying). Locked pairs skip steps 3–6.

Input → Output	DV	Why
090111 coffee, not roasted → 090111 coffee, not roasted	1, locked	Same HS6 code (self-pair)
090111 coffee, not roasted → 090121 coffee, roasted	1, locked	Same coffee, roasted
090111 coffee, not roasted → 090112 coffee, decaffeinated	1, locked	Same coffee, decaffeinated
090210 green tea, packs ≤ 3 kg → 090220 green tea, packs > 3 kg	1, locked	Same tea, different packing
520811 unbleached cotton plain weave → 520831 dyed cotton plain weave	1, locked	Same fabric, dyed

Boundary with siblings (step 4). "Very similar" means one can be turned into the other. Goods that differ in a fixed attribute — fibre, species, gender cut, base material — are siblings, not the same good: women's cotton jackets → women's synthetic jackets is not locked and goes to 0 at step 4. The direction must also be possible: dyed fabric → unbleached fabric is not locked and goes to 0 at step 5.

Size or format (30 October). When the same good comes in another size, lock only the feasible direction: wide → narrow and long → short film can be slit or cut. The reverse (narrow → wide film, a short roll of film → a long roll) is 0 at step 5. Intermediate goods such as paper and paperboard in rolls or sheets are not affected; they stay 1 as intermediate siblings (step 4). Endless belts of another circumference, or belts that differ in reinforcement, are different goods (0).

Step 3. Turn to 0: clearly irrelevant pairs

Rule: turn to 0 only when no route from step 6 could apply: the input would have to change into a different species, plant, fibre or substance, and it is not a constituent, feed, part or waste of the output. If in doubt, leave the pair at 1 for step 6.

Input → Output	DV	Why
200941 pineapple juice → 200410 frozen prepared potatoes	1→0	Different plant; juice is not an ingredient of potatoes
210500 ice cream → 210320 tomato ketchup	1→0	Unrelated food preparations
151110 crude palm oil → 151491 crude rapeseed oil	1→0	Refining cannot change the oil's source
710231 unworked diamonds → 711510 platinum catalysts	1→0	A stone does not become a metal
030562 salted cod → 030344 frozen bigeye tuna	1→0	Different fish species, and salted cod is not feed
030379 other frozen fish → 030332 frozen plaice	1→0	Different species; plaice is wild-caught, so no feed route
240120 unmanufactured tobacco → 621133 men's garments of man-made fibres	1→0	Different substance; tobacco is no constituent of a garment

Caution. Do not turn a pair to 0 at this step when the output is a mixture, a farmed animal or a waste product; leave it at 1 for step 6. Example: chicken cuts → 160249 (prepared pork and mixtures) looks irrelevant but is a mixture pair.

Fibres. Different synthetic fibres — nylon, polyester, acrylic, polypropylene — count as **one** fibre: nylon yarn → acrylic yarn is not a step-3 pair; it is an intermediate sibling (step 4) and stays 1. Different natural or artificial fibres (cotton → polyester, flax → jute, viscose → polyester) remain different fibres. **Blended fabrics (30 October):** fabrics of different main fibres that share a minor fibre, or where the input's main fibre is the output's minor fibre (polyester/cotton → viscose/cotton), are 1 with uncertainty.

Fish and other seafood. Different species are different goods: the pair is 0 at this step unless the input could feed the output (step 6, feed). The intermediate-sibling rule of step 4 does not apply to different species.

Process inputs (30 October)

A chemical, fuel, reductant or flux can be an input as a reagent, solvent, catalyst or binder. How specific the role must be:

Claim	DV	Example
A known production route of the output uses the input	1	290220 benzene → 290611 menthol (process solvent)
A specific role can be named (a named reaction or function), but no known route uses it	1, uncertain	Phenothiazine as antioxidant → caprolactam
Only a generic claim ("can act as a reagent", "fuel for process heat"), or both AI coders code 0, or the blind screen of both model families finds the one-sided 1 clearly invalid	1→0 (step 3)	292214 dextropropoxyphene → 293712 insulin

Parallel by-products (30 October)

Two parallel by-products of the same source, neither of which is made from or yields the other, are not inputs of each other: 020329 frozen pork → 050210 pig bristles is 0 (step 3). Bristles come from slaughter, not from processing pork. Waste routes that do stay 1 are in step 6.

Step 4. Turn to 0: downstream sibling products

Rule: both goods are **finished** products at the same stage, usually in the same heading or chapter, differing in a fixed attribute (fibre, species, cut, material), and neither physically goes into the other.

Input → Output	DV	Why
620432 women's cotton jackets → 620433 women's synthetic jackets	1→0	Parallel finished garments
620630 women's cotton blouses → 620640 women's man-made-fibre blouses	1→0	Parallel finished garments
640192 waterproof boots → 640110 waterproof boots with metal toe-cap	1→0	Parallel footwear
630622 synthetic-fibre tents → 630621 cotton tents	1→0	Parallel made-up articles
960820 felt-tip pens → 960810 ballpoint pens	1→0	Parallel pens

621490 shawls and scarves of other textiles → 620332 men's cotton jackets	1→0	A finished accessory, not a part of the garment

Exceptions that stay 1 (*route* = component), because one sibling is a part of the other:

Input → Output	DV	Why
640699 footwear parts → 640399 leather footwear	stays 1	Parts (uppers, soles) go into footwear
960891 nibs → 960839 fountain pens	stays 1	Nib is a pen component
960860 ballpoint refills → 960810 ballpoint pens	stays 1	Refill is a pen component
611780 knitted clothing accessories (trims, bows) → 610610 women's knitted cotton blouses	stays 1	Trims are attached to garments

Among clothing accessories, only trims and bows (611780, 621710) are components of garments. Finished accessories — handkerchiefs (6213), shawls and scarves (6214), ties (6215), gloves (6216), 611710 and 611720 — are siblings (0).

Intermediate siblings stay 1. Step 4 applies only when both goods are finished. Two different intermediate goods of the same material at the same stage stay 1 with *route* = sibling_intermediate. Intermediate goods are fibre, yarn (including sewing thread), fabric, steel and other metal semi-finished products, paper and paperboard in rolls or sheets, glass in the mass and semi-finished glass, feed residues, basic chemicals and parts. Fabrics of the same fibre but a different construction (woven, knitted, pile, nonwoven) are intermediate siblings. Garments, made-up textile articles, footwear, furniture, machines, vehicles, consumer articles and retail food preparations are finished. Exception (30 October): metal articles of the same metal (the articles of chapters 73–83; sheet piling, sections, rails and tubes of 7301–7306 are semi-finished forms) are not step-4 siblings, even when both are finished. Same heading: 1 (730810 bridges → 730840 scaffolding); another heading: 1 with uncertainty (table below). Other finished siblings stay 0.

Input → Output	DV	Why
720851 hot-rolled steel sheet → 721420 hot-rolled steel bars	stays 1	Intermediate steel products
480100 newsprint → 480411 kraft paper	stays 1	Intermediate paper grades
480591 other uncoated paper → 480512 straw fluting paper	stays 1	Intermediate paper grades
230230 wheat bran → 230400 soya-bean oil-cake	stays 1	Parallel feed residues
731815 screws and bolts → 731816 nuts	stays 1	Steel fastener parts of one heading
520811 cotton woven fabric → 600622 cotton knitted fabric	stays 1	Same fibre, different construction
540241 nylon filament yarn → 540243 polyester filament yarn	stays 1	Synthetic fibres count as one fibre
540110 synthetic sewing thread → 550959 synthetic staple yarn	stays 1	Thread and yarn of one fibre

How far "the same material" reaches (30 October)

Case	DV	Example

Semi-finished forms of one material (plates, bars, rods, wire, sections, sheet piling, rails, tubes, sheet glass, base paper, plastic sheets)	1	Steel plate → steel bars, as above
Parts of machines or vehicles, and metal articles: same heading and same material	1	730810 steel bridges → 730840 steel scaffolding
Different heading, linked only by the same base material; or same heading but a different material	1, uncertain	731815 steel bolts → 730511 welded line pipe; knitting-machine parts → calculator parts; road wheels → seat belts
Chemicals of one heading: the same compound in another grade or form	1	
Chemicals of one heading: a different compound	1, uncertain	Fatty alcohol → propylene glycol
Chemicals of one heading, when both AI coders code 0, or the blind screen of both model families finds the one-sided 1 clearly invalid	0	280120 iodine → 280110 chlorine

Metal articles of the same metal follow this table whatever route is claimed, including remelting (731441 steel wire netting → 730711 cast-iron tube fittings: 1, uncertain); a semi-finished form remelted into a metal article follows step 6 (730210 rails → 730711 cast-iron tube fittings: 0, made into a finished product). Where this table and the direct-remelting rule of step 6 ("made into a finished product = 0") overlap, this table applies (Professor Zhang, 1 November).

Complete goods inside other goods (30 October)

Case	DV	Example
A complete machine or appliance built into another finished good, where a real product combines them	1 (component)	Data transmission module → mobile phone
The same, but the combination is only conceivable	1, uncertain	Desk lamp → upholstered seat
A finished good cut or refitted into another good	1, uncertain	Fishing vessel → patrol vessel
A finished good taken apart for its parts	1→0 (step 5)	Car → car parts

Parts going into their own machine stay component 1, as before.

Step 5. Turn to 0: inverse (backward) pairs

Rule: the input is downstream of the output in the same production chain, so the real relationship runs the other way.

Input → Output	DV	Why
721123 cold-rolled steel → 720229 ferro-silicon	1→0	Ferro-silicon goes into steel, not the reverse
520811 cotton woven fabric → 520511 cotton yarn	1→0	Yarn is woven into fabric
410791 finished leather → 410120 raw bovine hides	1→0	Hides are tanned into leather

- Exception that stays 1: a reverse link through a waste or scrap code (step 6, recycling).
- Exception: direct remelting with no scrap code in between (step 6, *route* = recycle_direct), within the three levels of direct remelting in step 6.
- Taking a finished good apart for its parts is backward (0).
- Steps 2, 4 and 5 can largely be applied by rule (self-pairs and same-heading stage rules for step 2; sibling and backward rules for steps 4 and 5); AI coders only need to check the exceptions.

Step 6. Turn to 0 only where the expansive definition finds no route

Rule: every pair still at 1 is checked against the expansive routes below. It stays 1 if any route plausibly applies (record the route). It is turned to 0 only if you are sure no route exists — the cases listed in each table.

General principle. Use the broadest scope: keep 1 wherever **any** production route is technically possible, even if it is uncommon. A known route is a certain 1; a route that is possible but unconfirmed is 1 with uncertainty (decision rule above). If in doubt, keep 1.

Mixtures (route = mixture)

Input → Output	DV	Why
020714 frozen chicken cuts → 160249 prepared pork and mixtures	stays 1	Another meat in a meat mixture
070310 onions → 200570 prepared olives	stays 1	Flavouring or stuffing
070960 capsicum → 200570 prepared olives	stays 1	Pimento-stuffed olives
520811 unbleached cotton plain weave → 540752 dyed polyester filament woven fabric	1→0	“Mixed” refers to fibre content; a fabric → a fabric of another fibre, with no fibre in common, has no route (blends that share a fibre: step 3)

Food ingredients. An ingredient that is technically possible, even if uncommon, is 1 (linseed → peanut butter); one that is obviously never added is 0 (pineapple juice → frozen potatoes).

Minor constituents of textile goods (route = ingredient or mixture)

A garment, made-up article, yarn or fabric is classified by its main fibre, but it can contain other fibres: lining, pocketing, trim or blend. A fibre, yarn or waste of another fibre going into a yarn, fabric, garment or made-up article, or a fabric of another fibre going into a garment or made-up article, stays 1 (a fabric → a fabric of another fibre is not such a pair; see the mixtures table and the blended-fabric rule in step 3):

- **certain 1** when the input is **cotton or a man-made fibre** (synthetic or artificial). These are the usual blend partners and lining materials, and the HS classifies a mixed good by the fibre that predominates by weight, so a “cotton” garment can contain almost half man-made fibre;
- **1 with *uncertain* = 1** when the input is another fibre (wool, silk, flax, ramie, coarse animal hair, other textile materials), and when fabric of the **same fibre but another construction** goes into a garment (woven fabric → knitted garment and the reverse). These pairs count as 1 in DV.

Input → Output	DV	Why
540231 nylon filament yarn → 620342 men’s cotton trousers	stays 1	Man-made yarn as minor constituent

540752 dyed polyester woven fabric → 620311 men's wool suits	stays 1	Lining or blend
520821 cotton fabric → 630253 table linen of man-made fibres	stays 1	Cotton as blend or trim
540231 nylon filament yarn → 520851 printed cotton fabric	stays 1	Minor yarn in a fabric
510710 wool yarn → 610463 women's knitted synthetic trousers	1, uncertain	Wool blend: possible, less common
500720 silk woven fabric → 570110 wool carpets	1, uncertain	Silk as minor constituent
520811 cotton woven fabric → 610610 women's knitted cotton blouses	1, uncertain	Same fibre, other construction (woven trim)

Waste and scrap (route = recycle_waste)

Input → Output	DV	Why
551431 synthetic woven fabric → 550510 synthetic fibre waste	stays 1	Cutting waste
411200 finished sheep leather → 411520 leather parings and dust	stays 1	Shaving and buffing waste
510310 wool noils → 510521 combed wool in fragments	stays 1	Waste reprocessed into fibre
Yarn, fabric or fibre → 630900 worn clothing; fibre → 6310 rags	stays 1	Waste of goods made from the input (30 October)
851410 resistance furnaces → 854810 spent batteries	1, uncertain	Waste of a component of the input (30 October)
020329 frozen pork → 050210 pig bristles	1→0	Parallel by-product (step 3)

Direct remelting and re-processing (route = recycle_direct; convention of 30 October)

Remelting or re-processing with no scrap code in between is a link only within one material, and only as far as the following three levels:

Input → Output	DV	Level
721320 hot-rolled rod → 720610 non-alloy ingots	stays 1*	Back to the input's own upstream form
761010 aluminium doors and windows → 760110 unwrought aluminium	stays 1*	As above
780500 lead pipes → 780110 refined lead	stays 1*	As above
540752 dyed polyester woven fabric → 540233 textured polyester yarn	stays 1*	As above: yarn or fibre of the same fibre
700991 unframed mirrors → 700510 float glass	stays 1*	As above: the semi-finished form the article is made from
732111 steel stoves → 730110 steel sheet piling	1, uncertain*	Another primary or semi-finished form of the same material

731300 barbed wire → 730511 welded line pipe	1, uncertain*	As above
870323 passenger cars → 870790 truck bodies	1→0	Made into a finished product
680422 grinding wheels → 680520 abrasive paper	1→0	As above
400922 vulcanised rubber hose → 400239 synthetic rubber	1→0	Vulcanised rubber (4007–4017) cannot be remelted
392010 polyethylene film → 392061 polycarbonate sheet	1→0	Different polymers

- **Upstream forms** are ingots or unwrought metal, glass in the mass, pulp, primary plastics of the same polymer, the semi-finished form the article is made from, and yarn or fibre of the same fibre.
- **Glass** (cullet) and **paper** (repulping) qualify, and **thermoplastics** only for the same polymer (Professor Zhang, 30 October). Vulcanised rubber (4007–4017) is 0; reclaimed rubber (4003) and rubber waste (4004) remain possible outputs of rubber goods.
- **Extensions confirmed by Professor Zhang on 1 November:** (i) plastics that cannot be remelted are 0 (thermosets such as epoxide, alkyd, unsaturated polyester, amino and phenolic resins; silicones; cellulose and its derivatives; regenerated cellulose; vulcanised fibre); (ii) rubbers in primary forms only to the same polymer (natural rubber → synthetic rubber is 0); (iii) a code “of other plastics” counts as a polymer different from every named one; (iv) a laminate or component route does not create a link between different plastics; (v) vulcanised rubber ground into crumb and used as an ingredient of compounded or unvulcanised rubber (4005, 4006) is not remelting and stays 1 with uncertainty.
- Metal articles of the same metal are decided by the same-material table of step 4, which takes precedence over these levels.

* Direct remelting is 1 in the main DV and coded *route* = recycle_direct. The robustness variable DV_excl_remelt sets every recycle_direct HS2002 pair to 0.

Broad scope: wastes, raw materials and uncommon uses

Input → Output	DV	Rule
292212 diethanolamine → 382520 sewage sludge	stays 1	Any chemical (ch. 28–38) → wastes of heading 3825 (recycle_waste)
290322 trichloroethylene → 382541 halogenated waste solvents	stays 1	As above
252400 asbestos → 252330 aluminous cement	stays 1	Any mineral of ch. 25 → cement (ingredient)
251200 diatomite → 252329 Portland cement	stays 1	As above
681099 other concrete articles → 680710 asphalt articles	stays 1	A stone, concrete or ceramic good crushed into filler or aggregate (30 October)
251511 crude marble → 252620 talc	1, uncertain	An associated mineral recovered from an ore (30 October)
480451 kraft paper → 482320 filter paper	1, uncertain	Further use that is only technically possible (forward)

A further use is a certain 1 when a known route uses it, even if uncommon; when it is only technically possible, it is 1 with uncertainty (decision rule). A complete good built into another good follows the table in step 4.

Feed (route = feed)

For farmed animals other than fish and seafood (meat, eggs, milk), a feed use follows the general principle: a standard feed is 1, and a feed use that is possible but not established is 1 with uncertain = 1. When the output is a fish or other seafood, the input must be a feed-grade good — whole low-value fish (herring, sardines, mackerel, other fish), fish meat and meal, crustaceans and their meal (krill), molluscs — and the output a farmed carnivorous fish or crustacean (trout, salmon, eels, sea bass, flatfish, bluefin tuna, shrimps and prawns, crabs) or a product made from it. Wild-caught outputs (other tunas, skipjack, cod, herring, plaice) and grazers or filter feeders are 0. Where a feed use is possible but not established, the pair is 1 with *uncertain* = 1.

Input → Output	DV	Why
030264 fresh mackerel → 160411 prepared salmon	stays 1	Feed fish → farmed salmon
030371 frozen sardines → 030191 live trout	stays 1	Feed fish → farmed carnivore
030629 other crustaceans and crustacean meal → 030621 rock lobster	1, uncertain	Rock lobster is farmed on such feed in parts of Asia
030410 fresh fish fillets → 030613 frozen shrimp	1→0	Fillets are a human-food form, not feed
030350 frozen herring → 160414 canned tuna	1→0	Canned tuna is wild-caught
030710 oysters → 030332 frozen plaice	1→0	Oysters are high-value food, not feed
any fish → abalone, oysters or mussels	1→0	Grazers and filter feeders do not eat fish feed

Other routes

Input → Output	DV	Route
010639 live birds → 020890 other meat	stays 1	forward
120400 linseed → 200811 peanut butter	stays 1	ingredient
350211 dried egg albumin → 350219 liquid egg albumin	stays 1	reconstitute
290220 benzene → 290611 menthol (process solvent)	stays 1	ingredient (process; see step 3, process inputs)

Two AI coders and conflicts

- **Setup.** Two independent coders from different model families (Claude and Codex) receive the same rules (steps 2–6 with these examples) and the same pairs. Neither sees the other's answer.
- **Units.** To code tens of thousands of pairs, pairs are grouped by heading pair (input HS4 × output HS4). Each coder gives one default code per unit plus exceptions for single input codes, output codes or pairs, with the 5 most similar coded pairs as worked examples.
- **Question asked.** Each coder decides whether the pair should be turned from 1 to 0, and at which step, and returns {DV, route, step, one-sentence reason}. For a 1 it also returns the level: *known* or *possible* (decision rule).

- **Combining rule.** Both say 0: DV = 0. Both say 1: DV = 1, with uncertain = 1 if either coder rated the link only possible. If they disagree, the pair stays 1 with *uncertain* = 1, and both codes and reasons are kept.
- **Decided by rule.** Where a convention (last section) settles a pair, the pair is coded by that convention, with the uncertainty the convention gives. Conventions for whole classes — metal articles of the same metal (step 4) and the levels of direct remelting (step 6, for semi-finished goods and articles as inputs) — apply even when both coders said 0.
- **Screen of one-sided 1s.** A one-sided 1 that no convention settles is screened for clear invalidity: it fails step 3, is physically or chemically impossible, runs in the reverse direction, names only a generic process role, or is a parallel by-product. One model family screens every such pair; the other screens, blind, the pairs the first found invalid. Only pairs that both find clearly invalid become 0 (example: tobacco → garments). All others stay 1 with uncertainty; for chemicals, a 1 stays whenever a specific role can be named (Professor Zhang, 1 November).
- **Applying a new convention to pairs already coded.** A convention applied by rule can turn a 1 to 0. Those pairs are screened once for another specific route (a component, ingredient or forward use); if one exists, the pair is 1 with uncertainty. A pair the convention's own example decides is not reopened, and neither is any plastics → plastics pair (extension (iv) of step 6).
- **Reliability.** Report Cohen's kappa between the two coders. Then code a stratified sample of about 300 pairs across sources, steps and routes to estimate the accuracy of the final codes, weighting each stratum by its size, by pairs and by records, with 95% intervals. Without a domain expert, the sample is coded by two new AI sessions that see only the rules and the pairs (no earlier answer), and a person reviews every pair where they disagree with each other or with the final code. The true value of a sampled pair follows the conventions in force; when a convention changes, the previous version is re-scored on the same true values. Errors that form a group are corrected by rule in every PTA not yet delivered, not pair by pair; audit codes do not change a final code pair by pair. (One exception, in PTA 6–10 only: 6 sampled pairs that both auditors coded 0, in classes where Professor Zhang said such pairs are 0, were set to 0; accuracy is also reported counting them as errors.) Report it as an independent AI audit, not as expert hand-coding.

Step 7. Scale up

(a) Exact matches

Any pair already in the coded master list (same HS2002 pair after concordance) takes the existing DV, route and uncertain values, with source = exact_copy. The master list grows with every coded PTA. Exact copies are re-copied whenever the master labels are corrected, so a pair has the same code in every PTA.

(b) Train a model on the coded pairs

- Features: input and output chapters and headings; same-chapter and same-heading flags; flags from the step 2, 4 and 5 rules; text embeddings of both descriptions and their interaction.
- Target: whether the pair stays 1 or is turned to 0 (optionally also route and step).
- Validation: hold out whole chapter pairs (grouped cross-validation) so the model cannot memorise near-duplicate pairs; random cross-validation overstates accuracy.

(c) Apply with confidence bands

Predicted probability	Action

≥ 0.9	Keep at 1 (source = ml)
≤ 0.1	Turn to 0 (source = ml)
Between 0.1 and 0.9	Send through steps 2–6 with the two AI coders, giving them the 5 most similar coded pairs as worked examples

Support condition. Use the bands only where the chapter pair has more than 30 coded pairs in the master list, and only for pair types where the audit (d) shows the bands are reliable. Other pairs go to the AI coders or to explicit rules. So far the bands have been used only for textile pairs confirmed by the textile rules.

(d) Audit

Check a random sample of model-coded pairs from each band against the two AI coders (or a person). If accuracy in a band falls below the threshold, narrow that band and send more pairs to the AI coders. When the model and an explicit rule disagree, the rule is applied where it finds a route; otherwise the pair goes to the AI coders.

(e) Explicit rules for structured chapters

Textile chapters 50–63 are coded by explicit rules. Each code has a stage (fibre, yarn, fabric, trim, made-up or garment, waste), a fibre and, for fabrics and garments, a construction (knitted or not). The rules apply steps 2–6 and the conventions above: forward stage with the same fibre = 1; minor constituents as in step 6; finished siblings = 0; backward = 0; intermediate siblings of one fibre = 1; sewing thread and trims → articles = component; waste reprocessed = 1; yarn, fabric or fibre → worn clothing (6309) and fibre → rags (6310) = 1. The textile rules do not yet code the blended fabrics of step 3 (different main fibres sharing a minor fibre, 1 with uncertainty); they are being extended before the next agreements are coded. In PTA 6–10 every such pair was a copy.

Chapter 59. Wicks (5908) and textile hosepiping (5909) → transmission belting (5910) = 0; coated or rubberised fabric (5901, 5903, 5906, 5907) → belting = 1; 5901 → tyre cord fabric (5902) = 1.

Reading “of other textile materials”. In the HS this means every fibre that is **not named** in the same group of subheadings. 610990 T-shirts “of other textile materials” include synthetic T-shirts, because only cotton has its own subheading (610910), so 540231 nylon yarn → 610990 is a main-fibre (forward) pair. 620449 dresses of other textile materials exclude wool, cotton, synthetic and artificial fibres, which have their own subheadings.

> **Example.** Another agreement restricts 030342 frozen yellowfin tuna → 160414 prepared tuna; the exact pair is already coded, so it is copied and stays 1 (forward). A new pair, 731815 steel bolts → 730511 welded line pipe, goes to the two AI coders; both find only the shared material, so by the same-material table it is 1 with uncertain = 1. A third pair, 230120 fishmeal → 020714 chicken cuts: one coder names fishmeal as poultry feed, the other codes 0; the screen does not find the claim clearly invalid, so it stays DV = 1 with uncertain = 1.

Summary of scope conventions

Question	Convention	Set
Main variable	DV. No variant counting only certain links (DV_strict) is computed	1 Nov
Overall principle	Known route or standard practice: 1. Possible but unconfirmed: 1 with uncertain = 1. Impossible or reverse: 0	30 Oct
Different intermediate goods of the same material at the same stage	1, route sibling_intermediate; finished siblings stay 0 (step 4)	19 Oct
Parts and metal articles; chemicals of one heading	Same heading and material (metal articles even when finished), or same compound: 1. Another heading but the same base material, the same heading but another material, or another compound of the same heading: 1 with uncertain = 1. Chemicals of one heading: 0 when both AI coders code 0 or both screens find the 1 clearly invalid. Metal articles: this rule takes precedence over the remelting levels	30 Oct / 1 Nov
Different synthetic fibres (nylon, polyester, acrylic, polypropylene)	One fibre	19 Oct
Fabrics of the same fibre, different construction	1 (sibling_intermediate)	19 Oct
Fabrics of different main fibres sharing a minor fibre	1 with uncertain = 1	30 Oct
Sewing thread ↔ yarn of the same fibre	1 (sibling_intermediate; thread → twine forward)	19 Oct
Clothing accessories → garments	Trims and bows: 1 (component). Finished accessories (scarves, ties, gloves): 0	19 / 28 Oct
Any chemical (ch. 28–38) → wastes of heading 3825	1 (recycle_waste)	19 Oct
Any mineral of ch. 25 → cement (2523)	1 (ingredient)	19 Oct
Process inputs (reagent, solvent, catalyst, binder, fuel)	Known route: 1. Specific role: 1 with uncertain = 1 (also in the screen of one-sided 1s). Generic claim only: 0	30 Oct / 1 Nov
Further use	Known route, even if uncommon: 1. Technically possible but unconfirmed: 1 with uncertain = 1	19 / 30 Oct
Minor constituent of another fibre in a garment, made-up, yarn or fabric	Cotton or man-made input: 1. Other fibres, or the same fibre in another construction (fabric → garment): 1 with uncertain = 1	19 Oct

Direct remelting with no scrap code in between	Own upstream form: 1. Other primary or semi-finished form of the same material: 1 with uncertain = 1. Finished product: 0. Vulcanised rubber: 0. Glass and paper qualify; thermoplastics only the same polymer. Non-meltable plastics 0, rubbers only the same polymer, “other plastics” a polymer of its own, no laminate route between plastics; vulcanised rubber crumb as an ingredient: 1 with uncertain = 1. Route recycle_direct; 0 in DV_excl_remelt	19 / 30 Oct / 1 Nov
A complete machine or appliance inside another good	Real combination: 1. Conceivable only: 1 with uncertain = 1. Not applied to pairs both coders already coded 0	30 Oct / 1 Nov
A finished good cut or refitted into another good	1 with uncertain = 1. Taken apart for its parts: 0	30 / 28 Oct
Waste	Waste of goods made from the input (6309, 6310): 1. Waste of a component of the input: 1 with uncertain = 1. Parallel by-product of the same source: 0	30 Oct
Minerals and stone	Associated mineral recovered from an ore: 1 with uncertain = 1. Crushed into filler or aggregate: 1	30 Oct
Food ingredients	Technically possible, even if uncommon: 1. Obviously never added: 0	19 / 30 Oct
The same good in another size or format	Only the feasible direction is the same good; the reverse is 0	30 Oct
Fish and other seafood of different species	0 (step 3), unless the input could feed the output: feed (1); possible but unestablished feed: 1 with uncertain = 1	19 Oct
“Of other textile materials”	Every fibre not named in the same group of subheadings	19 Oct
Chapter 59	Wicks, hosepiping → belting: 0. Coated fabric → belting, 5901 → tyre cord fabric: 1	28 Oct
Inputs “WO” or “any”	Rows deleted	28 Oct
The two AI coders disagree and no convention settles it	1 with uncertain = 1, unless both model families find the 1 clearly invalid (0)	19 / 28 Oct
Codes already delivered	Not re-judged when a later convention is adopted; copies keep them. Uncertain flags of blended fabrics are updated at the next reissue of PTA 1–5	30 Oct / 1 Nov